

# Georges Parfait Djimefo Kapen | Data Scientist | AI Agent Developer

Montréal, Québec, Canada

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## Profile

Data scientist with expertise in **AI agents**, **Large Language Models**, **advanced analytics**, and **machine learning** applied to **telecom and networking** environments. PhD expected Summer 2026. Proven ability to design, deploy, and evaluate **production-grade LLM-based agentic systems** including RAG pipelines, orchestration frameworks, and autonomous decision engines on AWS. Strong foundation in statistical modeling, data engineering, and cross-functional collaboration.

## Technical Skills

**LLM / Agents:** LangChain, LangGraph, LangSmith, MCP, RAG (FAISS), CRAG, RAGAS, Claude (Haiku, Opus), Kiro, Prompt Engineering

**AI / ML:** PyTorch, TensorFlow, Scikit-Learn, GNN, DQN, MARL, NLP, Computer Vision, MLflow

**Data:** PySpark, Databricks, Snowflake, Delta Lake, Pandas, SQL, Relational Databases, ETL/ELT

**Cloud:** AWS (SageMaker, Bedrock, EKS, Lambda, CloudWatch, S3, ECR, CloudFormation), Kubernetes, Docker

**Languages:** Python, Java, C++, SQL, R, JavaScript, TypeScript

**DevOps:** CI/CD (GitHub Actions, CodeBuild, OIDC), Git, Jenkins, GitLab, Terraform

**Visualization:** Streamlit, D3.js, Plotly, Dash, Power BI, Tableau

**Methods:** Agile/Scrum, TDD, Statistical Modeling, MLOps, Infrastructure as Code, Design Patterns

## Relevant Experience

### Ericsson

Montréal

*Machine Learning Intern — AI Agents & Telecom Analytics*

2026

- Designed and deployed a **production RAG Autonomous Agentic AI system** for telecom radio unit log anomaly detection, serving Ericsson's enterprise clients.
- Trained a custom **Dual-Attention SLM** (522M params) achieving **F1 90.6%**, 99.97% accuracy, and **1 ms/log** inference latency on AWS SageMaker.
- Built a **6-tool ReAct agent** orchestrated by Claude Haiku 4.5 on Bedrock: SageMaker scoring, FAISS RAG (631 patterns), RF physics world model, pattern classification, CloudWatch health monitoring, and on-prem EricAI.
- Implemented **self-correcting RAG (CRAG)** with quality grading, **RAGAS** faithfulness/relevancy evaluation, and evidence-based **meta-cognition** confidence scoring (HIGH/MEDIUM/LOW).
- Defined and applied **evaluation metrics** for LLM-based systems: latency, faithfulness, relevancy, confidence calibration, and business-impact scoring.
- Engineered a **production-scale map-reduce pipeline**: triage → cluster by failure signature → parallel agent investigation → adaptive cross-cluster synthesis. Handles 120+ anomalies in ~12 min.
- Deployed the agent as a **K8s pod on EKS** exposed via Cloudflare tunnel with a Streamlit UI accessible from any device on the corporate network.
- Built an **industry-standard event pipeline**: S3 upload → Lambda → SQS → K8s worker → results in S3 → SNS email alerts on critical anomalies.
- Designed a **fully autonomous retraining loop**: drift-triggered + weekly retraining via EventBridge, with auto-promotion only when the new model's F1 exceeds the current production model.
- Implemented a **7-stage CI/CD pipeline** (GitHub Actions + AWS CodeBuild with OIDC): test → OWASP security scan → CloudFormation infra → staging gate → deploy → smoke test → auto-rollback.
- Stack:** Python, AWS (SageMaker, Bedrock, EKS, Lambda, SQS, SNS, S3, ECR, CloudFormation, CloudWatch), LangChain, LangGraph, LangSmith, FAISS, MCP, Claude, Kiro, Docker, Kubernetes, CI/CD, Terraform.

### Intact Financial Corporation

Montréal

*AI Developer I — Data Science & Analytics*

2022–2024

- Analyzed and processed **large-scale datasets** to uncover patterns and deliver actionable insights.
- Monitored ML pipelines and developed **automated testing** solutions for data quality assurance.
- Partnered with cross-functional stakeholders to translate requirements into scalable data solutions.
- Stack:** PySpark, SQL, Databricks, Snowflake, AWS, Shell, Git.

### Intact Financial Corporation

Montréal

*Data Engineering Intern — Advanced Analytics*

2022

- Built and maintained data pipelines processing large-scale insurance datasets (PySpark, SQL, Databricks, Snowflake, AWS).

## Additional Experience

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| <b>Polytechnique Montréal</b><br><i>Lecturer</i> <ul style="list-style-type: none"><li>Teach the Python procedural programming course.</li></ul>                                          | <b>Montréal</b><br>2025–2026 |
| <b>Polytechnique Montréal</b><br><i>Lecturer</i> <ul style="list-style-type: none"><li>Teach the Python procedural programming course.</li></ul>                                          | <b>Montréal</b><br>2022–2023 |
| <b>Université de Montréal</b><br><i>Scientific Assistant — MOOC AI</i> <ul style="list-style-type: none"><li>Design courses and practical work in supervised learning (Python).</li></ul> | <b>Montréal</b><br>2021–2022 |
| <b>Ministry of Tourism and Recreation</b><br><i>Contractual Statistical Economist Engineer</i> <ul style="list-style-type: none"><li>Analyze complex datasets</li></ul>                   | <b>Cameroon</b><br>2012–2014 |

## Selected Projects

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| <b>Polytechnique Montréal</b><br><i>NLP Project</i> <ul style="list-style-type: none"><li>“Extraction of keywords from texts as well as their type” (Python).</li></ul>                                                                                                                                                  | <b>Montréal</b><br>2021      |
| <b>Polytechnique Montréal</b><br><i>Data Visualization Project — Chaire de recherche en fiscalité et en finances publiques</i> <ul style="list-style-type: none"><li>“Develop a web application to compare taxes and provincial and federal tax rates by year.”</li><li>Implement a mockup with Plotly.js, D3.</li></ul> | <b>Montréal</b><br>2021      |
| <b>Polytechnique Montréal</b><br><i>Research Initiation — Deep Learning</i> <ul style="list-style-type: none"><li>“Modeling of Voice Recognition in Android systems to secure bank payments.”</li><li>Model with Deep Learning (Python); implement and deploy on Android mobile (Java).</li></ul>                        | <b>Montréal</b><br>2020–2021 |
| <b>FLEX GROUP</b><br><i>Programming Intern — OCR</i> <ul style="list-style-type: none"><li>“Modeling of Automatic Optical Character Recognition.”</li><li>Implement models in Python, PHP.</li></ul>                                                                                                                     | <b>Laval</b><br>2019–2020    |

## Publications

- [1] *Energy-Efficient and Near Real-Time Routing Algorithm for Disaster Monitoring in Wireless Sensor Networks*  
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
  - Designed **GAPF**, a novel **meta-learning** framework fusing MARL policies via a **GCN encoder** with only **1 473 parameters**.
  - Achieved  $\geq 98\%$  packet delivery,  $2\text{--}3.5\times$  less energy,  $13\times$  less memory than baselines.
- [2] *Carbon-Aware Autonomous Data Reduction and Self-Healing Routing for 6G-Integrated Disaster Sensor Networks*  
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
  - Proposed **CALASH**, the **first protocol** jointly optimizing Lifecycle Carbon Intensity via a hybrid **Lyapunov–DQN** engine and **MAPE-K** self-healing over 6G.
  - Achieved 60% longer network lifetime and 33% lower LCI than baselines.
- [3] *Zero-Trust Shielded Graph-Structured Decision Control for Secure Autonomous Recovery in AI-Native 6G Networks*  
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
  - Designed **CASTER-ZT**, the **first framework** combining **GNN-based** decision making with a formally bounded **zero-trust** admissibility shield (10 proven properties).
  - Achieved 74–98% rogue detection with **zero false positives**, 3 ms latency within O-RAN 10 ms budget.

## Education

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Polytechnique Montréal</b><br><i>PhD in Computer Engineering</i> <ul style="list-style-type: none"><li>AI-driven routing, carbon-aware networking, and zero-trust security for 6G sensor networks.</li></ul> | <b>Montréal, Canada</b><br>Summer 2026 |
| <b>Polytechnique Montréal</b><br><i>Engineering Diploma in Software Engineering</i> <ul style="list-style-type: none"><li>Concentration: Artificial Intelligence — Data Sciences.</li></ul>                     | <b>Montréal, Canada</b>                |
| <b>Advanced National School of Engineering (ENSPY)</b><br><i>Master's Degree in Applied Statistics</i> <ul style="list-style-type: none"><li>Equivalency: Master's in Probability — Statistics.</li></ul>       | <b>Yaoundé, Cameroon</b>               |
| <b>University of Yaoundé I</b><br><i>Master's Degree in Mathematics</i>                                                                                                                                         | <b>Yaoundé, Cameroon</b>               |

- Equivalency: Specialized Higher Studies Diploma (DESS) in Mathematics.

## Licenses & Certifications

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- AWS Certified Cloud Practitioner (CLF-C01)  
*Certification Prep Path (Pluralsight), Montréal*
- AWS Certified Developer — Associate (DVA-C02)  
*Certification Prep Path (Pluralsight), Montréal*

## Conferences & Seminars

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### **PolyCongrès 2026, Polytechnique Montréal**

**Montréal, Canada**

*Oral Presentation*

*Mar–Apr 2026*

“Energy-Efficient and Near Real-Time Routing for Disaster Monitoring in Wireless Sensor Networks — Graph-Attentive Policy Fusion (GAPF).” Georges Parfait Djimefo, Ranwa Al Mallah, Samuel Pierre.

### **20th Conf. African Assoc. of Insect Scientists**

**Yaoundé, Cameroon**

*Oral Presentation*

*Oct 2013*

Djimefo P., Tchunte M. & Le Gall P. — “Biogeography of communities of insects using clustering methods.”

### **29th Annual Conf. Int. Society for Clinical Biostatistics**

**Copenhagen, Denmark**

*Poster*

*Aug 2008*

“HIV incidence estimation from repeated cross-sectional surveys: a comparative study of parametric and non-parametric approaches by simulation.”

## Volunteering

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- Volunteer at the IEEE branch of Polytechnique Montréal, Montréal, 2018.
- Volunteer at the ConFoo conference, Montréal, 2018.